

# Stage 16N Report

Updated 2026-06-22

## 1 Overview

Stage 16N investigates restart-preserved cycle-jump behavior for the Chaboche UMAT Abaqus benchmark. The R4H diagnostic tested whether source-split restart files differ from restart records produced by long restarted replay runs. The R4I-R recovery rerun then repeated the cycle-280 source-split branch with durable lightweight copy-back and absolute reference CSVs.

## 2 Results

R4H completed six PBS/Abaqus jobs successfully with `Exit_status=0`. Long-replay restart records passed exactly: R4H1 at cycle 500, R4H3 at cycle 500, and R4H5 at cycle 750 all had zero global, local, and diagnostic S11 error. Short source-split restart cases remained inconsistent: R4H2 failed at cycle 500 with 11.83033% max primary local error, R4H4 was review-level at cycle 500 with 9.3860617%, and R4H6 was review-level at cycle 750 with 8.0369449%.

The R4I-R first wave finished by 2026-06-20 13:28 CEST and was absent from the live PBS queue on 2026-06-21 05:47 CEST. PBS history no longer returned records for jobs 1350601 and 1350602, so the current classification uses scratch case status files, PBS stdout, Abaqus completion messages, and copied comparison CSVs. R4I-R1 passed exactly at cycle 500; R4I-R2 failed with the same source-split signature as R4H2.

| Case | Mode                            | Endpoint | Status | Global %   | Primary local % | S11 %      |
|------|---------------------------------|----------|--------|------------|-----------------|------------|
| R4H1 | Long replay 280–500             | 500      | pass   | 0          | 0               | 0          |
| R4H2 | Source 250–281, restart 280–500 | 500      | fail   | 1.1887403  | 11.83033        | 0.92936729 |
| R4H3 | Long replay 270–500             | 500      | pass   | 0          | 0               | 0          |
| R4H4 | Source 250–271, restart 270–500 | 500      | review | 2.5314199  | 9.3860617       | 1.2392968  |
| R4H5 | Long replay 505–750             | 750      | pass   | 0          | 0               | 0          |
| R4H6 | Source 500–506, restart 505–750 | 750      | review | 0.31896796 | 8.0369449       | 1.5702038  |

### R4I-R first-wave recovery

| Case   | Source solve       | Restart | Endpoint | Status | Global %  | Primary local % | S11 %      |
|--------|--------------------|---------|----------|--------|-----------|-----------------|------------|
| R4I-R1 | Deck-clone 250–281 | 280–500 | 500      | pass   | 0         | 0               | 0          |
| R4I-R2 | Buffered 250–300   | 280–500 | 500      | fail   | 1.1887403 | 11.83033        | 0.92936729 |

R4I-R1 shows that cloning/truncating the clean direct replay deck shape repairs the cycle-280 branch. R4I-R2 shows that a longer buffered generated source alone does not repair that branch; the dominant mismatch remains `HOLE_RING_SDV8_MAX`.

## R4I-R second wave and deck-clone confirmations

R4I-R3 and R4I-R4 were submitted on 2026-06-21 through the guarded storage-gated scratch workflow. Both jobs requested one node with 16 cores, one MPI rank, 16 OpenMP threads, 90 GB memory, and 24 h walltime. R4I-R3 reproduced the generated-buffer review signature at restart 270 with 9.3860617% maximum primary local error. R4I-R4 reproduced the generated-buffer review signature at restart 505 with 8.0369449% maximum primary local error.

R4I-R5 then confirmed the deck-clone/truncate strategy for the 270–500 branch: deck-clone source 250–271, restart 270–500, zero global, primary-local, and S11 error at cycle 500. R4I-R6 and R4I-R6B did not classify the 505–750 branch scientifically. R4I-R6 failed during continuation `.stt` writing near cycle 731, and R4I-R6B failed earlier during source `.stt` writing near cycle 503. These are infrastructure/output-writing failures, not numerical method failures.

## R4K controller preparation

A 2026-06-22 storage/source audit showed `/scratch9` at 99% use and found no visible clean cycle-505 restart source under `/scratch9/pr21vyci`, `/scratch/pr21vyci`, or the home project clone. The next prepared work is therefore split into two controllers: R4K1 runs the confirmed 250-branch exact deck-clone control `250 -> exact 280 -> 500`; R4K2 performs a 505-branch restart-source preflight and stops before Abaqus solve if no valid source is found. Both controllers include phase timing and lightweight evidence copy-back.

The controllers were submitted on 2026-06-22 through the guarded wrapper. R4K1, job `1350764.mmaste02`, finished with `Exit_status=0` after 06:01:15 walltime and 20:39:54 CPU time. Abaqus completed successfully, and the exact-control comparison at cycle 500 passed with zero maximum global error, zero maximum primary-local error, and zero diagnostic S11 error. This confirms the 250-branch deck-clone exact control `250 -> exact 280 -> 500`. The corrected R4K2 preflight is job `1350767.mmaste02`; it finished successfully in 6 s walltime and found one large non-R4I candidate from the earlier R4E exact-target scratch area, `stage16n_r4e2_exact_500_to_505_solve_506_to_750_exact_target_source`, with a 4.9 GB `.stt` and successful cycle-505 source `.sta`.

The follow-up storage-light validation controller R4K2B, job `1352353.mmaste02`, used that preserved R4E2 cycle-505 candidate directly. It did not generate a new source `.stt`; the scratch case linked the preserved source files, read `STEP=505`, `INC=54`, and continued cycles 506–750 with no continuation restart-write request. PBS recorded `Exit_status=0`, `Stageout_status=1`, 05:41:43 walltime, 19:16:27 CPU time, and about 3.38 average active cores out of 16. Abaqus completed through cycle 750, but the comparison was nonzero: 0.31896796% maximum global error, 8.0369449% maximum primary-local error, and 1.5702038% diagnostic S11 error. Therefore R4K2B is a scientific validation failure of the R4E2 505 candidate, not an infrastructure failure. The stage-out warning is recorded separately because lightweight evidence was recovered and committed.

The 250-cycle restart branch is now validated for deck-clone/truncate source construction, including the exact-control case R4K1 with zero global and local comparison error. In contrast, the 505-cycle branch remains unresolved. The preserved R4E2 cycle-505 restart candidate completed the R4K2B continuation but reproduced the previous review signature, with 8.0369449% maximum primary-local error. Therefore, the R4E2 candidate is not accepted as a validated 505 -> 750 continuation source, and further 505-branch work is postponed to avoid additional large restart-file generation. The next work should focus on storage-light true-jump candidates on the validated 250 branch.

## R4L prepared storage-light true-jump controller

The next prepared production controller is `stage16n_r4l_250branch_storage_light_controller`. It uses the validated 250-branch deck-clone/truncate source construction and keeps the continuation decks restart-read-only. The controller runs R4L1 first, targeting the conservative 250 -> 270 -> 500 true-jump case, then runs R4L2 around 250 -> 280 -> 500 only if R4L1 passes. If R4L1 reviews or fails, R4L2 is not run; instead the controller writes diagnostic summaries for source-cycle state, first solved continuation cycle, comparison details, and local hole-ring metric ranking. R4L copies only lightweight evidence and deletes case-local heavy Abaqus files after classification. After confirming no active jobs and `/scratch9/pr21vyci` at about 2.7 TB, the controller was submitted as PBS job `1353907.mmaste02` on 2026-06-22 and started on `mnode079`.

| Case  | PBS job | Endpoint | Status | Global %   | Primary % | local S11 % |
|-------|---------|----------|--------|------------|-----------|-------------|
| R4K1  | 1350764 | 500      | pass   | 0          | 0         | 0           |
| R4K2B | 1352353 | 750      | review | 0.31896796 | 8.0369449 | 1.5702038   |

### 3 Images

No R4H images were generated in this update.

### 4 Tables

The key R4H and R4I-R comparison tables are included in the Results section. Lightweight R4I-R evidence is stored under:

```
runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/
stage16n_restart_control/r4ir_restart_source_buffer_recovery/
```

Prepared R4K controller files are stored under:

```
runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/
stage16n_restart_control/r4k_deck_clone_controllers/
```

R4K1 lightweight result evidence is stored under:

```
runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/
stage16n_restart_control/r4k_deck_clone_controllers/
R4K1_250branch.controller/
```

R4K2B lightweight validation evidence is stored under:

```
runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/
stage16n_restart_control/r4k_deck_clone_controllers/
R4K2B_505candidate.validation.controller/
```

Prepared R4L controller files are stored under:

```
runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/
stage16n_restart_control/r4l_250branch_storage_light_controller/
R4L_250branch_storage_light.controller/
```

The earlier R4H CSV evidence is stored under:

```
runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/
stage16n_restart_control/r4h_restart_source_diagnostics/
```

### 5 Failures and Debugging

No R4H job failed at PBS/Abaqus level. The original R4I comparison failed because remote reference CSV paths were missing after extraction, and its scratch outputs were not recoverable. R4I-R fixed that workflow by staging absolute reference CSVs and copying lightweight evidence before comparison. The scientific failure now narrows to generated source-split consistency: R4I-R2 reproduces the R4H2 error even with a larger 250–300 source buffer, and R4I-R3/R4I-R4 show that generated post-target buffering does not repair the 270 or 505 branches. R4I-R6/R4I-R6B additionally exposed scratch `.stt` write failures on the 505 branch.

## 6 Conclusion and Next Steps

R4H, R4I-R, and R4K1 show that long-replay/deck-clone restart records are clean on the confirmed 250 branch, while generated short-source decks remain non-equivalent even with post-target buffering. Deck-clone/truncate is confirmed for 250–270–500 and for the exact-control path 250–280–500. The 505–750 branch now has a completed R4K2B validation of the preserved R4E2 cycle-505 candidate, and that candidate failed scientifically with the same nonzero 750-cycle signature observed in earlier source-split/review cases. R4J9/R4J10 remain blocked. The next useful solver path is the prepared R4L storage-light 250-branch true-jump controller; the 505 branch is parked until a new exact-control gate or redesign is defined.